

Table 1. Red HYDRA LED Power Efficiency Test Results

Iteration n	V in (Volt)	I in (Ampere)	V out (Volt)	I out (Ampere)	Wout (W)	Wout (W)	%
1	19.2	0.093	18.82	0.093	1.786	1.75	98.02083
2	22.5	0.139	19.37	0.152	3.128	2.944	94.14037
3	26	0.121	19.4	0.151	3.146	2.929	93.11507
4	30	0.112	19.48	0.156	3.36	3.039	90.44286
5	31	0.108	19.5	0.157	3.348	3.062	91.44265
Average Error Rate					2.953	2.953	2.953

Table 2. Blue HYDRA LED Power Efficiency Test Results

Iteration n	V in (Volt)	I in (Ampere)	V out (Volt)	I out (Ampere)	Wout (W)	Wout (W)	%
1	28.40	0.081	28.30	0.081	2.300	2.292	99.64789
2	31.60	0.144	29.47	0.151	4.550	4.450	97.79294
Average Error Rate					3.425	3.371	98.72041

Table 3. White HYDRA LED Power Efficiency Test Results

Iteration n	V in (Volt)	I in (Ampere)	V out (Volt)	I out (Ampere)	Wout (W)	Wout (W)	%
1	28.4	0.045	28.37	0.044	1.278	1.248	97.67449
2	31.6	0.16	31.17	0.162	5.056	5.05	99.87223
Average Error Rate					3.167	3.167	3.167

Table 4. Comparison of Power Efficiency Test Results with Previous Research

Color	Difference in Average Power Efficiency		
	Lotus (%)	Hydra (%)	Difference (%)
Red	92.69	93.43236	0.7424
Blue	94.23	98.72041	4.4904
White	96.22	98.77336	2.5534



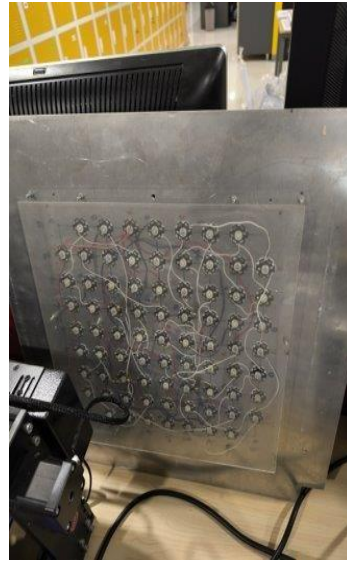
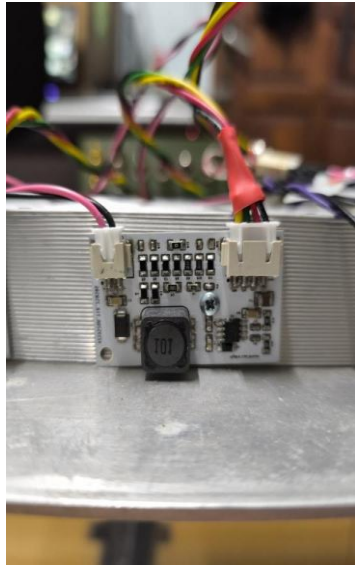


Fig. 3. Hydra System Overview